

Evaluating data augmentation techniques in the classification of multi-station seismic-volcanic signals

Franz Yupanqui Machaca , Pablo Salazar Reinoso , Claudio Meneses Villegas 

Abstract—Effective volcano monitoring depends on quickly identifying seismic events to better grasp activity levels and issue timely warnings. Conventional approaches rely on manual analysis, which can lead to bias and demands considerable effort. This work seeks to characterize several data augmentation techniques in terms of similarity and diversity metrics, and evaluate their impact on the performance of classification of seismic-volcanic signatures. These techniques are used to generate diverse datasets on which seismic-volcanic event classification models are trained, to analyze which techniques provide the best classification performance. A primary objective of using data augmentation is to balance the dataset, ensuring greater diversity in the event feature space, and maintaining a minimal similarity with the original data. The generated datasets are categorized into two types: (1) balanced datasets composed of real and augmented data, and (2) balanced datasets composed of only augmented data. Furthermore, applying similarity and diversity metrics to the augmented data reveals that the fully augmented datasets exhibit greater diversity than the real datasets. It is also observed that several of these augmented datasets yield superior performance in the classification models.

Index Terms—Deep Learning, Transfer Learning, Data Augmentation, Machine Learning classification, Seismic-volcanic signals, FID, KID, MS-SSIM.

I. INTRODUCTION

A. The problem of recognition

The signatures are not dangerous; however, volcanic and seismic events may be dangerous and occur unexpectedly, although they present some signals before they occur. Globally, these events are continuously registered using seismometers. Afterward, expert geophysicists classify these data. Seismic-volcanic activity has been studied for many years. Currently, this type of activity is registered by sensors that record signals in three dimensions. The seismic-volcanic pattern recognition process is typically done manually, although there are state-of-the-art studies that perform the process in a semi-automatic or automatic manner [1]–[6]. These studies regularly consider one of the three available channels, one of several stations, and one volcano. In this context, classical machine learning (ML) and deep learning (DL) algorithms

applied to the automatic classification of seismic-volcanic signatures are becoming an emerging research field to overcome the limitations of manual classification.

This study proposes and applies a series of data augmentation (DA) techniques to volcanic seismic events and analyzes their impact on the performance of event classification models. A database of events collected from the Lascar volcano over an 8-month period and labelled by CKELAR-VOLCANES experts was considered. Five classes (Table II) were included according to the classification given by CKELAR-VOLCANES; see also [1]. Given the nature of the seismic phenomena that occur in and around the volcano, the database is unbalanced, requiring a data balancing technique; therefore, DA techniques were considered. DA techniques were applied to generate two types of datasets: real and augmented data that balance the dataset and datasets with fully augmented data. Similarity and diversity metrics were applied to the data generated by the DA techniques to analyze these characteristics. Classification models were trained using DL and Transfer Learning (TL) techniques. The models were evaluated using precision, accuracy, recall, F1 score, and confusion matrix metrics.

Each seismic event is represented by its respective spectrogram, so that these images can be processed. Using the Transfer Learning strategy, event classification models were trained, considering the pre-trained AlexNet deep artificial neural network. Event classification models are trained on different datasets generated by different Data Augmentation techniques, and their performance is reviewed.

The results of the experiments are presented in terms of performance metrics, comparing the impact of each data augmentation method on the classification of seismic-volcanic events. The evaluation included models trained on real and augmented data, as well as models trained on entirely augmented datasets. The influence of similarity and diversity metrics on model generalization was also analyzed.

The remainder of this paper is organized as follows. The most recent bibliography on this subject is presented in Section II. Section III presents the methodology. The methods used are described in Section IV. The main results are presented in Section V. Section VI discusses the results of this study, and conclusions are presented in Section VII.

II. LITERATURE REVIEW

A. A summary of methods used in classification of volcanic seismic events

The seismic-volcanic signals are time series that are not stationary and contain information about the physical processes occurring in the volcanic system. Magma movement, gas

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TABLE I: Current status of seismic-volcanic signal classification studies.

Models and presentation year	Volcano - Country	Examples	Spectrogram	Channels	Stations	Classes	Accuracy (%)
Hidden-Markov-Model-Based FSE(2013) [4]	Galeras – Colombia	300	No	–	2	3	96
Hidden-Markov-Model-Based FSE(2013) [4]	Galeras – Colombia	400	No	–	2	4	85.2
Topic Model (VT, LP, TR)(2015) [7]	Galeras – Colombia	300	No	–	1	3	94.1
Topic Model (VT, LP, TR, HB)(2015) [7]	Galeras – Colombia	400	No	–	1	4	83.8
Bag-of-Words (VT, LP, TR)(2015) [7]	Galeras – Colombia	300	No	–	1	3	91.9
Bag-of-Words (VT, LP, TR, HB)(2015) [7]	Galeras – Colombia	400	No	–	1	4	81.8
Support Vector Machines(2018) [3]	Ubinas – Perú	109,609	No	–	1	6	93.5±0.5
CNN LeCun(2018) [8]	Llaima – Chile	15,895	Yes	–	1	4	97.0
GRU-RNN y LFB(2018) [9]	Isla Decepción – Antártida	2,193	No	–	1	5	91.6*
LSTM-RNN y LFB(2018) [9]	Isla Decepción – Antártida	2,193	No	–	1	5	93.8*
Gaussian Mixture Model + uFilter +5f (2019) [10]	Cotopaxi – Ecuador	668	No	–	1	2	96.7
Bayesian Neural Network (2019) [11]	St. Helens – EE.UU.	25,301	No	–	3	3	95.7
Bayesian Neural Network (2019) [11]	Bezymianny – Russia	24,916	No	–	3	3	94.5
Bayesian Neural Network (2019) [11]	St. Helens – EE.UU., Bezymianny – Russia	50,217	No	–	6	3	94.6
Bayesian Neural Network (2019) [11]	St. Helens – EE.UU., Bezymianny – Russia	50,217	No	–	6	6	92.1
Support Vector Machines (2020) [6]	Ubinas – Perú	3,792	No	3	2	5	83.2
DCNN1(2020) [2]	Cotopaxi – Ecuador	1,108	Yes	–	1	2	95
CNN LeNet(2020) [12]	Colima – México	9,332	Yes	–	1	7	88.3
Deep Gaussian Processes - DGP4(2020) [13]	Colima – México	9,332	No	–	–	7	94.8
BIRCH k=2(2020) [14]	Cotopaxi – Ecuador	668	No	–	1	2	88
DNN SeismicNet(2020) [15]	Llaima – Chile	3,592	No	–	1	4	95
STuM y MPCA-DR(2021) [16]	Ubinas – Perú	5,136	No	3	2	5	91.9*
TCN en b-E3(2021) [17]	Bezymianny – Rusia	3,844	Yes	–	1	3	94
RNN-LSTM(2022) [18]	Isla Decepción – Antártida	2,193	No	3	–	5	84.9
ResNet152 – TL(2022) [19]	Lascar – Chile	13,430	Yes	1	10	5	93.2
AlexNet – TL Artificial(2022) [19]	Lascar – Chile	67,095	Yes	1	10	5	98.9
AlexNet – TL 10-folds(2022) [19]	Lascar – Chile	13,430	Yes	1	10	5	89.1
AlexNet – DL CAE3 Reconstructed Event(2022) [19]	Lascar – Chile	13,430	Yes	1	10	5	91.2
AlexNet – TL Real+Artificial 20Hz(2023) [1]	Lascar – Chile	13,430	Yes	1	10	5	91.5
AlexNet – TL Artificial 20Hz(2023) [1]	Lascar – Chile	67,095	Yes	1	10	5	99.4
AlexNet – TL Artificial 20Hz, Real(Test)(2023) [1]	Lascar – Chile	6,145	Yes	1	10	5	99.5
CNN M1-Exp1(2023) [20]	Nevados del Chillán – Chile	313,951	Yes	–	5	8	92.2
UNet & Semantic Segmentation(2024) [21]	Nevados del Chillán Volcanic Complex, Laguna del Maule, Villarrica and Puyehue-Cordón - Chile	24,493	No	1	8	5	91.0**

* Correct classification rate (CCR).

** F1 Score.

emissions, and rock fracturing are just a few of the sources that generate these signals. Understanding volcanic activity and assessing eruption risks requires the analysis of these signals. In the analysis and treatment of seismic-volcanic time signals, transformations are usually applied to represent the time domain observations in terms of the frequency and cepstral domains, and using statistical and information theory metrics, among others [3], [6]. Among the methods used for the analysis of this type of signal, the most common are the Hidden Markov model (HMM) [4], [22], [23] with their improvements, such as the Hidden Semi-Markov Models (HSMM), the Continuous Wavelet Transform (CWT) [22] in the detection of singularities, Gaussian Mixture Model (GMM) [24], and others, such as the HMM-based generative embedding [4].

Currently, there are several works that propose the automatic classification of seismic-volcanic signals that consider supervised Machine Learning (ML) methods such as Support Vector Machines (SVM) [3], [5], [6], [24]–[27]; Multilayer Perceptron(MLP) [6], [24], [25]; Random Forest [3], [6], [12]; Linear Discriminant Analysis [6]; Convolutional Neural Networks (CNN) [2], [8], [12]; Temporal Convolutional Neural Network(TCN) [12], [17]; Bayesian Neural Networks [11]; Tensor Learning Framework with Multilinear Principal Component

Analysis(MPCA) and Support Tensor Machine(STM) [16] techniques on multichannel and multistation data; Scalable Variational Inference for Gaussian Process (SVGP) and Deep Gaussian Processes (DGP) [13]; Bag of Words in seismic document classification as a set of seismic words [7].

Unsupervised methods include Kohonen Self-Organizing Map (SOM) [24], [28], Cluster Analysis(CA) [24], Balanced iterative reducing and clustering using hierarchies(BIRCH) [14]. Transfer Learning (TL) was applied in the classification of volcanic seismic events using CNN [12], [19].

Deep Learning (DL) has been applied to seismic-volcanic signal classification, experimenting with different CNN [19] and Recurrent Neural Networks(RNN) architectures [2], [15], [20], and through Deep Gaussian Processes (DGP) [13].

RNNs provide real-time tracking capabilities for volcanic activity, even if the seismic sources change over time. It has been applied to seismic-volcanic signal classification and tested several architectures [2]. In detection and classification, the following RNN architectures have been applied: vanilla-RNN, Long Short-Term Memory (LSTM), and Gated-Recurrent Unit (GRU) [9]; in [17] vanilla-RNN and LSTM, in addition to identifying the degree of specialization of the neurons.

An Autoencoder (AE) is an excellent unsupervised learning algorithm. The AE is considered a nonlinear compression

structure that reduces the signal size and dimensionality reduction towards another latent dimension. In [19] various AE architectures generated reconstructed and encoded seismic events, which were classified using different DL and TL model architectures. In [17] AEs were applied to obtain the segmentation mask of a seismic event.

DA involves generating new valid data from existing data following certain rules and restrictions. Amplitude deformation, frequency deformation, noise addition, and horizontal shifts were applied to the seismic signals [8]. In [1] the importance of DA in avoiding biased models in the classification of seismic-volcanic events was shown.

In reference to volcanic seismic data, most studies consider a single station channel for the classification of volcanic seismic events, but other studies consider several channels and stations [5], [6], [20], [29]. However, [12] and [20] showed that using a multi-station scheme is more precise than a single-station approach. This state-of-the-art review in seismic-volcanic signal classification is summarized in Table I.

III. METHODOLOGY

Currently, Machine Learning is used for the recognition of seismic-volcanic signal patterns. In fact, our previous research developed [1], [19] models that implemented a new methodology for classifying these types of seismic signals using Deep Learning, Transfer Learning, and spectrograms. The rotation technique was used as a data augmentation technique to avoid data imbalance. Current research on recognizing seismic-volcanic signal patterns considers data from a single Chilean volcano. The Millennium Institute on Volcanic Risk Research – Ckelar Volcanoes¹ monitors many of these volcanoes using seismic stations equipped with triaxial sensors.

A. Lascar volcano

The Lascar volcano is located in northern Chile (23°22' S, 67°44' W; 5592 m a.s.l.), 270 km NE from Antofagasta and 70 km SE from San Pedro de Atacama. CKELAR-VOLCANES has been monitoring the area using a network of 11 three-component seismic stations. Short-period 2 Hz seismometers were continuously monitored at 200 Hz. A signal database was built using only the Z channel. The stations are distributed around the volcano, as shown in Figure 1.

B. Lascar's database

The Lascar database contains 6,145 seismic events classified by geophysicist experts belonging to CKELAR-VOLCANES. The data correspond to the period between March and October 2018. The tables II and III show the data distribution.

IV. METHODS

We proposed a solution based on transfer learning because this helps to speed up training and improve the performance of deep learning models [11], [12], in addition to the limited number of examples available for training the model from

scratch. For TL, the pre-trained AlexNet [31] model was considered. The input data were spectrograms of the data labeled (waveforms) for each seismic event class. The processing sequence consists of five steps.

- 1) describing the pre-process applied to the time series (earthquake);
- 2) detailing the generation of spectrograms from the labeled earthquakes;
- 3) using data augmentation techniques to increase the amount of data in the training and test datasets;
- 4) constructing the prediction model; and, finally,
- 5) estimating the performance of the model.

In the following paragraphs, each step is explained.

- 1) Pre-processing: The seismometers record data at 200Hz, 500Hz or 100Hz. We resampled each event at 100 Hz. The data were centered at the mean. A Butterworth Highpass filter was applied at 1 Hz. Then, we applied a Butterworth Bandpass filter at frequencies between 1 Hz and 10 Hz. The next step was to normalize the data. This pre-processing task was performed using the ObsPy Python toolbox [32].
- 2) The spectrograms were calculated applying a short-time Fourier transform with formula

$$S[x(t)](n, k) = \left| \sum_{m=0}^{N-1} x(m) \cdot w(m-n) \cdot e^{i2\pi mk} \right|^2 \quad (1)$$

where $x(t)$ and $y(t)$ represent the seismic signal and short-time Fourier transform sliding window, respectively. The sliding window size was set to 1 with a 95% overlap. The generated spectrograms are RGB images of 224×224 pixels.

- 3) Data augmentation techniques were considered to avoid unbalanced datasets and prevent biased models. Many data augmentation techniques were implemented, which are presented in the next section.
- 4) The pre-trained model AlexNet [31] was considered for retraining over spectrograms, using the scheme of transfer learning.
- 5) The model performance was evaluated using metrics described in the following sections. The metrics are implemented on the TorchMetrics Python toolbox².

Codes for reproducing our method are available at.³

A. Data Similarity

Kullback–Leibler divergence (KL divergence) has been used to measure similarities between synthetic and real datasets, and is defined as

$$D_{KL}(P||Q) = \sum_i P(x_i) \cdot \log \left(\frac{P(x_i)}{Q(x_i)} \right) \quad (2)$$

where P and Q are the probability distributions whose distance is calculated over the samples x_i of the distribution [33]. Kullback–Leibler divergence (KL divergence) is not a symmetric

¹<https://ckelar.org>

²<https://lightning.ai/docs/torchmetrics/stable>

³<https://github.com/franznet/daseismicsignals>

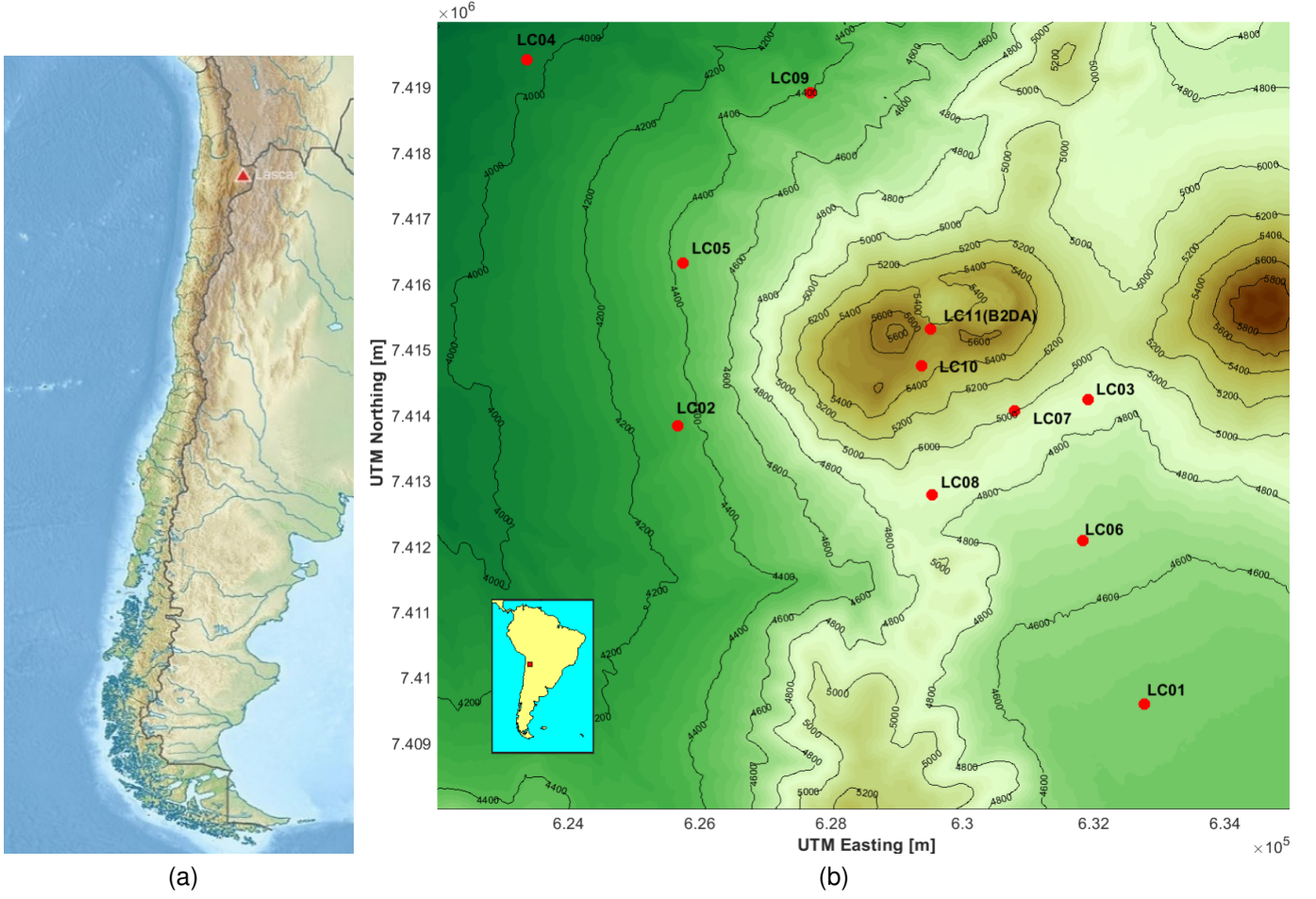


Fig. 1: a) Location map of Lascar volcano [30]. b) Red dots indicate the positions of the seismic stations.

TABLE II: Contribution of events from each station to the Lascar database.

Event\Station	LC01	LC02	LC03	LC04	LC05	LC06	LC07	LC08	LC09	LC10	B2DA	Total
HY	0	0	0	7	45	4	0	3	12	136	6	213
LP	11	0	4	2	142	28	5	60	23	195	107	577
TC	9	0	11	97	284	36	157	127	144	1,171	162	2,198
TR	1	0	5	29	21	4	61	13	35	292	10	471
VT	2	0	18	52	84	6	206	14	41	2,249	14	2,686
TOTAL	23	0	38	187	576	78	429	217	255	4,043	299	6,145

TABLE III: Statistics of the events belonging to the Lascar volcano.

Events	Quantity (%)	Quantity	Maximum (s)	Minimum (s)	Mean (s)	Median (s)	Mode (s)	Standard Deviation (s)
HY	3%	213	265	10	57.2	59	40	35.3
LP	9%	577	600	9	82.3	67	25	63.5
TC	36%	2,198	670	9	78.5	65	20	56.2
TR	8%	471	216	9	46.2	31	20	33.8
VT	44%	2,686	189	5	19.6	17	10	12.3
TOTAL	100%	6,145	670	5	49.9	28.0	20	50.0

distance, so it can be symmetrized to give rise to the so-called Jensen–Shannon divergence (JS divergence), defined as

$$JS(P||Q) = D_{KL}(P||(P+Q)/2) + D_{KL}(Q||(P+Q)/2) \quad (3)$$

A measurement to quantify the distance between time series distributions. It is based on the calculation of the Wasserstein distance between time-series data. This metric is defined by measuring the Wasserstein distance of the energy between frequencies. The probability distributions have a Wasserstein–Fourier distance, which is calculated as follows [33]:

$$WF([x], [y]) = W_2(s_x, s_y) \quad (4)$$

where s_x and s_y are the normalized power spectral densities of the distributions.

B. Data Augmentation

1) *Rotation*: This transformation considers the rotation of the spectrogram around the time axis within a given percentage range of the signal length and at a given axis position. The axis position can be at the beginning, middle, or end of a spectrogram. When applying this technique, the combination of the percentage rotation value, location of the rotation axis, and spectrogram of the event to which it will be applied is obtained. When generating new instances, care was taken to ensure that these three values were not repeated, thus ensuring unique instances in the dataset (Figure 2).

2) *Jittering*: Jittering is a data augmentation technique commonly employed in time series analysis [33], [34]. It is considered one of the simplest yet most effective transformation-based methods, as it involves adding noise to a time series. Formally, an original time series $x = x_1, \dots, x_t, \dots, x_T$ is transformed into $x' = x_1 + \epsilon_1, \dots, x_t + \epsilon_t, \dots, x_T + \epsilon_T$ by introducing noise at each time step t , where ϵ typically represents Gaussian noise distributed as $N(\theta, \sigma^2)$. The standard deviation σ of the added noise is a hyperparameter that must be defined in advance [34]. This approach assumes that the time series patterns of a dataset are inherently noisy. Such cases may include data derived from sensors, audio signals, or electroencephalograms (EEG). For instance, jittering has been applied to wearable sensor data for Parkinson’s disease monitoring (Figure 3).

3) *Drifting*: One challenge in both online and offline settings is data drift, defined as a change in the data distribution over time. A random walk is often used to simulate such drift. In mathematics, a random walk is classified as a stochastic process that represents a path composed of successive random steps in a mathematical space, such as the integers or the reals. A simple example is the integer number line $\mathbb{Z}$, where the walk begins at θ and moves either $+1$ or -1 with equal probability at each step. More generally, a random move of $+a$ or $-a$ with equal probability can occur in any vector space, such as the real space, where $a \in \mathbb{R}$ denotes the step magnitude [35]. Let d define a path starting at position $d_1 = \phi(\tau)$. A random walk can then be modeled as:

$$d_t = d_{t-1} + \phi(\tau) \quad (5)$$

where ϕ is the random variable describing the probability distribution for the next step, and τ is the time interval between consecutive steps. Consequently, an original time series $x = x_1, \dots, x_t, \dots, x_T$ can be transformed into $x' = x_1 + d_1, \dots, x_t + d_t, \dots, x_T + d_T$. Data drift simulation can be used to generate more robust training datasets, thereby enhancing model resilience and reducing sensitivity to drift when it arises in real-world data (Figure 4).

4) *Genetic Algorithms*: Genetic Algorithms (GA) belong to the broader class of Evolutionary Algorithms (EA) and are inspired by the principle of natural selection. Their primary objective is to generate high-quality solutions to complex problems using bio-inspired operators such as selection, crossover, and mutation [36]. In [37], GA have been applied to data augmentation of time series derived from centers of pressure for the detection of neuropathies. Building on GA operators, a seismogram data augmentation technique was developed with the following procedure:

- *Selection*: Real seismograms of the same class are selected, without repetition, to serve as parents.
- *Crossover*: Genetic information from two or more parent seismograms is combined to generate a new instance. This is achieved by exchanging time series segments between the parents’ chromosomes.
- *Mutation*: Small modifications are introduced in a limited portion of the population. In this case, a smoothing operation is applied to the scales of the exchanged segments.
- *Validation*: KL and JS divergences are used to measure similarity between datasets. Values closer to zero for both KL and JS divergences indicate greater similarity in the frequency-domain power distribution between the original and augmented seismic signals.

Two strategies were considered for the crossover operation:

- *Without time adjustment*: The segments exchanged between parents may differ in length, resulting in a child seismogram with a duration that can vary significantly from that of the original seismograms (Figure 5).
- *With time adjustment*: The exchanged segments are rescaled to equal length by applying a time-scaling transformation, ensuring that the generated seismogram maintains a duration comparable to the original signals (Figure 6).

5) *SpecAugment*: SpecAugment is a data augmentation technique designed for Automatic Speech Recognition (ASR). Its primary purpose is to help the network learn useful features that are robust to time warping, to the partial loss of frequency information, and to the partial loss of small signal segments. It is a simple and computationally inexpensive method, as it acts directly on the log-mel spectrogram as if it were an image, allowing for online application during training [38]. SpecAugment policies consist of three main components: frequency masking, time masking, and time warping [39], [40]. In this work, we apply SpecAugment on the spectrograms using frequency masking, time masking, and time warping policies (Figure 7).

6) *Interpolation AEs*: Autoencoders can interpolate in certain situations by decoding the convex combination of the

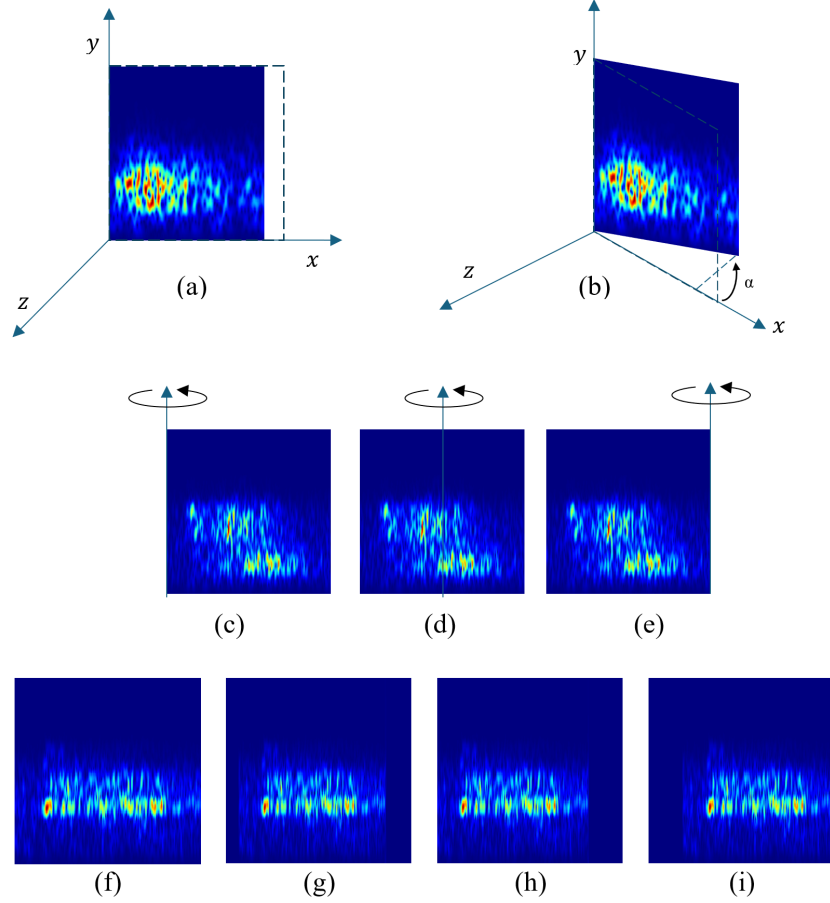


Fig. 2: Transformation by rotations, the time stretching (a) is produced when we apply the angle of rotation (b); The possible rotation axes are located at the beginning (c), center (d), and end (e) of the spectrogram, respectively. Application examples include: (f) original spectrogram, (g) 19% rotation around the central axis, (h) 23% rotation around the initial axis, and (i) 24% rotation around the final axis.

latent codes of two data points, thereby producing an output that semantically blends the features of the data points [41]. First, an input $x \in \mathbb{R}^{d_x}$ is passed through an encoder $z = f_\theta(x)$, parameterized by θ , to obtain a latent code $z \in \mathbb{R}^{d_z}$. The latent code is then passed through a decoder $\hat{x} = g_\phi(z)$, parameterized by ϕ , to produce an approximate reconstruction $\hat{x} \in \mathbb{R}^{d_x}$ of the input x . We consider the case in which both f_θ and g_ϕ are implemented as multi-layer neural networks. The encoder and decoder are trained jointly with respect to θ and ϕ to minimize a distance measure between the input x and the output $\hat{x}$, such as the squared L_2 distance $|x - \hat{x}|^2$. Interpolation with an autoencoder refers to the process of using the decoder g_ϕ to reconstruct a mixture of two latent codes. Typically, the latent codes are combined via a convex combination, so that interpolation amounts to computing $x_\alpha = g_\phi(\alpha z_1 + (1 - \alpha)z_2)$ for some $\alpha \in [0, 1]$, where $z_1 = f_\theta(x_1)$ and $z_2 = f_\theta(x_2)$ are the latent codes corresponding to the data points x_1 and x_2 (Figure 8).

C. Model performance - Metrics

Machine learning metrics make it possible to quantify the performance of a trained model and to compare it with other generated models or with the current baseline. Evaluating the

TABLE IV: Confusion matrix.

		Predicted	
		Positive (P)	Negative (N)
Actual	Positive (P)	True positive (TP)	False negative (FN)
	Negative (N)	False positive (FP)	True negative (TN)

learning algorithm is a crucial part of any project. While classification accuracy is most commonly used to assess model performance, it is insufficient on its own to provide a complete evaluation. Therefore, various complementary evaluation metrics are considered. Among the different classification metrics, we can mention:

- *Confusion Matrix*: A tabular representation of true labels versus model predictions. Although not a performance metric in itself, it serves as the foundation for deriving many other evaluation measures (Table IV). Where TP denotes true positives, TN true negatives, FP false positives, and FN false negatives.
- *Recall (Sensitivity)*: Measures the proportion of actual positive cases that the model correctly identifies.

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

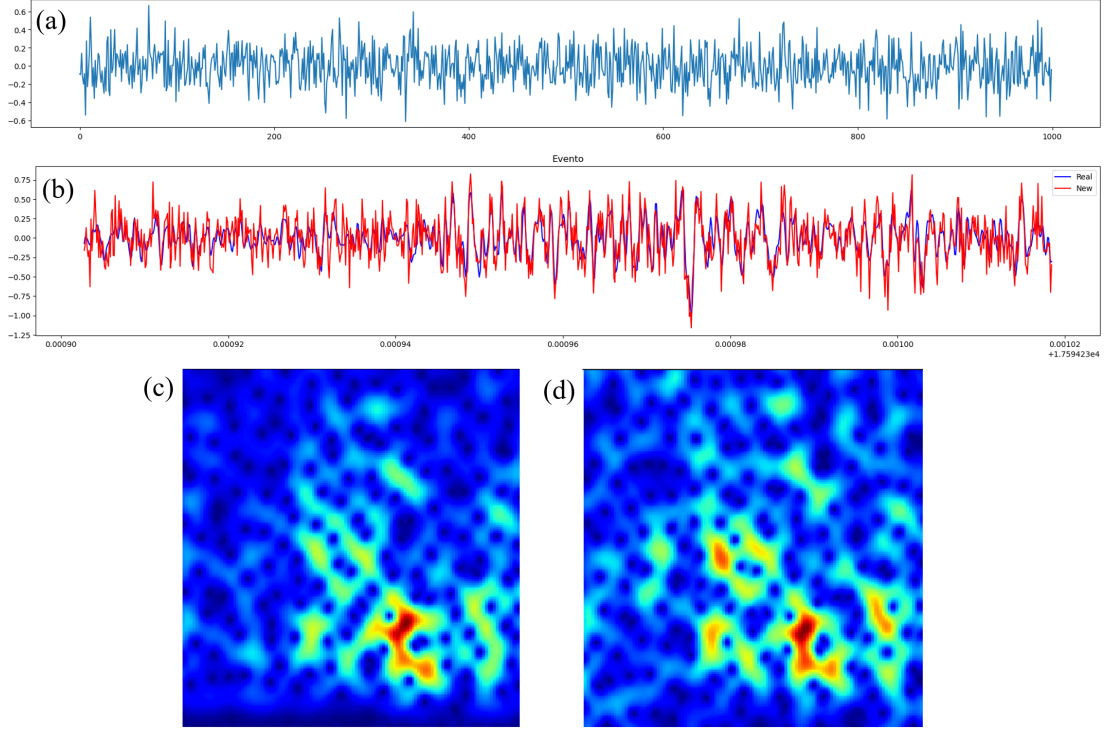


Fig. 3: Generation of a new signal using jittering. (a) Gaussian noise with distribution $N(\mu = 0, \sigma = 0.2)$. (b) Original (blue) and generated (red) signals. (c) Spectrogram of the original signal, and (d) spectrogram of the jittered signal.

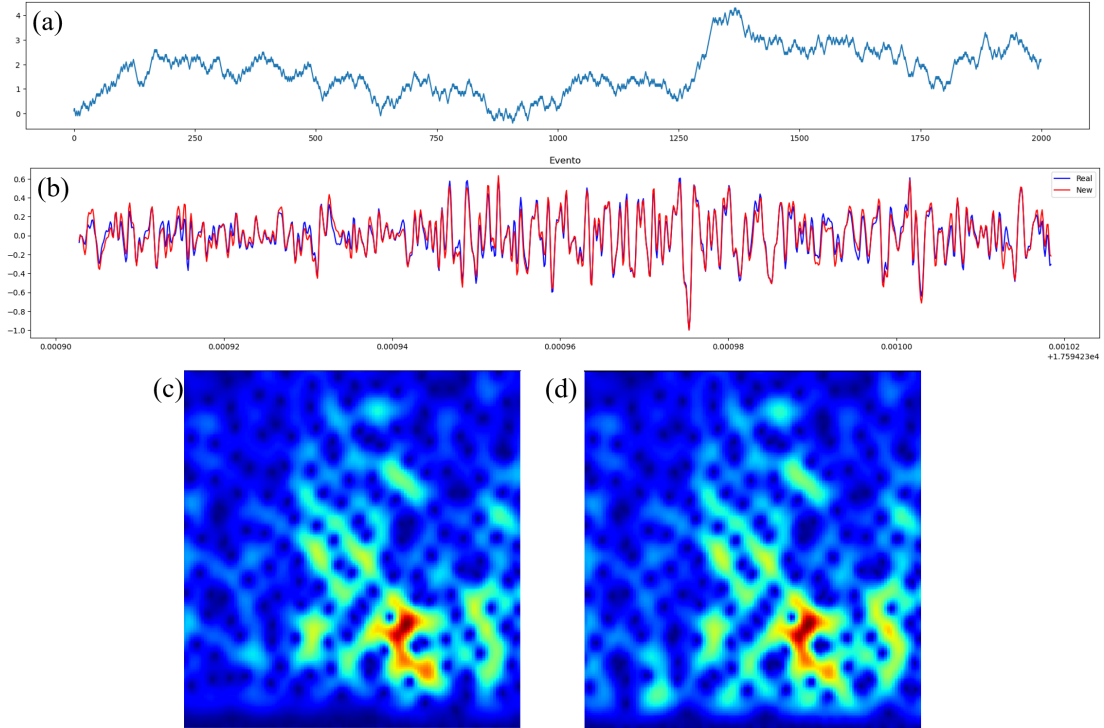


Fig. 4: Generation of a new signal using drifting. (a) Drifting signal with $\tau = 0.2$. (b) Original signal (blue) and generated signal (red). (c) Spectrogram of the original signal, and (d) spectrogram of the drifted signal.

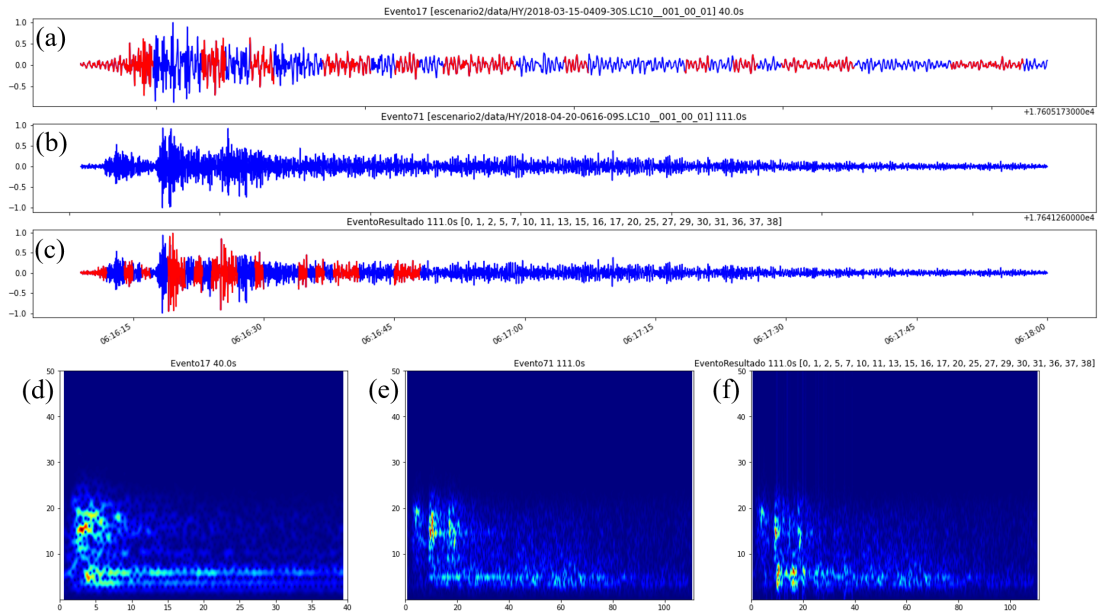


Fig. 5: Generation of a new signal using a Genetic Algorithm without time adjustment. (a) and (b) show the parent signals; the red color highlights the exchanged segments. (c) The resulting signal generated by the Genetic Algorithm. (d) and (e) show the spectrograms of the parent signals. (f) Spectrogram of the generated signal.

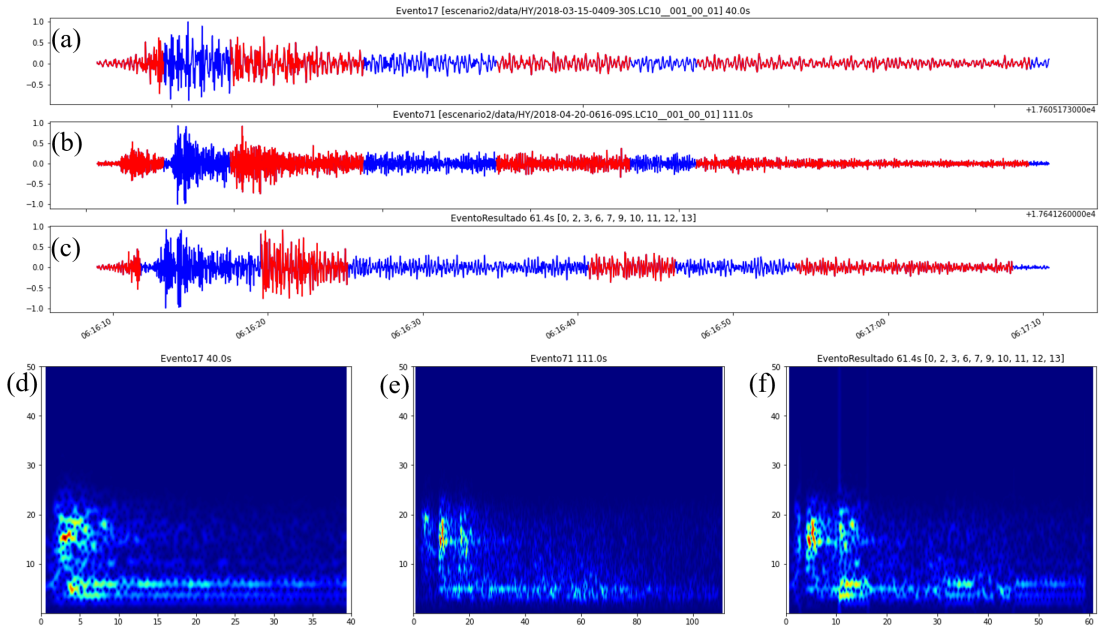


Fig. 6: Generation of a new signal using a Genetic Algorithm with time adjustment. (a) and (b) show the parent signals; the red color highlights the exchanged signals. (c) The resulting signal generated by the Genetic Algorithm. (d) and (e) display the spectrograms of the parent signals. (f) Spectrogram of the generated signal.

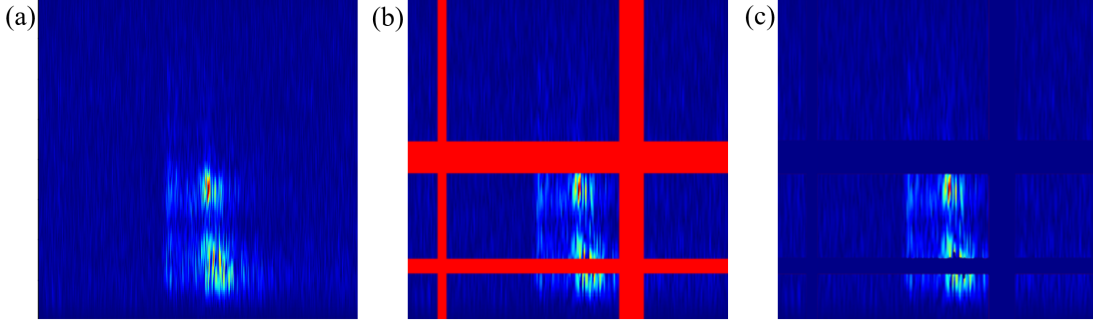


Fig. 7: Generating new signal with SpecAugment. (a) Original spectrogram. Spectrogram with two time-frequency distortions. The distortions are shown in red (b), and with white noise in (c). Generation of a new signal using SpecAugment. (a) The original spectrogram. (b) Spectrogram displaying two time-frequency distortions (highlighted in red). (c) Spectrogram displaying the same distortions with white noise.

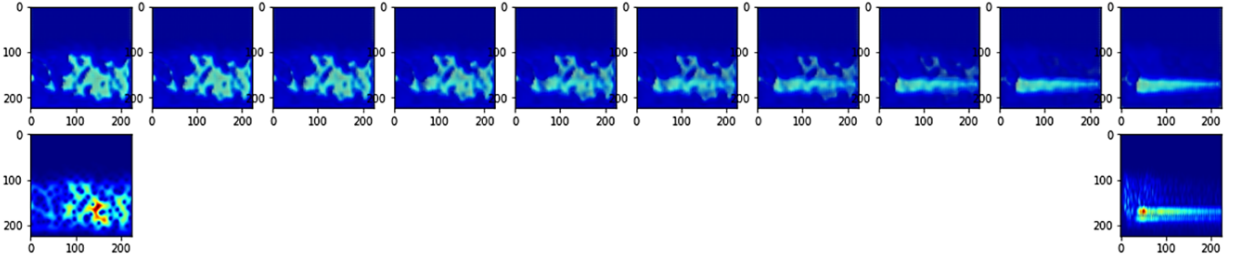


Fig. 8: Interpolating events (HY) at different intensity levels yields greater diversity.

- *Precision*: Measures the proportion of positive predictions that are correct, indicating the model’s reliability in classification tasks.

$$Precision = \frac{TP}{TP + FP} \quad (7)$$

- F_β : Combines precision and recall into a single value, facilitating comparison across models. The parameter $\beta > 0$ adjusts the relative importance of recall versus precision: values $\beta > 1$ favor recall, while $\beta < 1$ favor precision. This metric is particularly useful for imbalanced datasets.

$$F_\beta = (1 + \beta^2) \cdot \frac{precision \cdot recall}{\beta^2 \cdot precision + recall} \quad (8)$$

- *Accuracy*: Measures the overall proportion of correctly classified cases. However, it can be misleading when dealing with imbalanced datasets.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

- *Specificity*: Measures the proportion of actual negative cases that the model correctly identifies.

$$Specificity = \frac{TN}{TN + FP} \quad (10)$$

- *Cohen’s kappa*: Measures the agreement between your model and human evaluators, taking randomness into account. It is particularly useful for imbalanced datasets.

$$\kappa = \frac{p_0 - p_e}{1 - p_e} \quad (11)$$

where p_0 is the empirical probability of agreement and p_e is the expected agreement when both annotators assign labels randomly⁴.

D. Evaluation metrics for data augmentation

Data augmentation techniques have become a widely adopted strategy for improving the generalization capacity of deep neural networks. An important consideration in this context is the balance between similarity and diversity in the data [42]. Similarity reflects the degree of resemblance between the original and augmented data, whereas diversity captures the variation in complexity introduced through augmentation. These metrics were calculated using the framework Python TorchMetrics⁵ de Lightning AI, the creators of PyTorch Lightning.

1) *Fréchet Inception Distance (FID)*: The Fréchet Inception Distance (FID) is a widely used metric for evaluating the quality of images generated by Generative Adversarial Networks (GANs) [43]. Its primary objective is to quantify the similarity between generated and real images [44]. Compared to earlier metrics such as the Inception Score (IS), the FID offers an improvement because it leverages statistics from real-world samples for comparison, whereas the IS does not [45]. The core idea of the FID is to embed both real and generated images into a vision-relevant feature space and then compute the distance between the distributions of these embeddings. In practice, this feature space is typically the penultimate layer

⁴https://lightning.ai/docs/torchmetrics/stable/classification/cohen_kappa.html

⁵<https://lightning.ai/docs/torchmetrics/stable>

(pool3, with 2048 features) of an Inception-V3 classifier pre-trained on ImageNet [45].

$$FID(\mu_r, \Sigma_r, \mu_g, \Sigma_g) = \|\mu_r - \mu_g\|_2^2 + Tr(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{\frac{1}{2}}) \quad (12)$$

where (μ_r, Σ_r) and (μ_g, Σ_g) denote the sample mean and covariance of the embeddings of the real and generated data, respectively, and $Tr(\cdot)$ indicates the matrix trace [45]. The theoretical range of the FID is $[0, +\infty)$, as it is a distance measure. Lower values correspond to higher-quality images [46]. In theory, an FID of 0 indicates that the feature distributions of the generated and real images are identical in terms of mean and covariance under the Gaussian approximation [44]. Empirical studies have shown that FID correlates well with human judgments of image fidelity [44].

2) *Kernel Inception Distance (KID)*: The squared Maximum Mean Discrepancy (MMD) is used by the Kernel Inception Distance (KID) to measure the difference between the feature representations of real and generated images. These features are extracted from the Inception network. Unlike the Fréchet Inception Distance (FID) [44], KID employs an unbiased estimator, which makes it more reliable, particularly when the number of test images is smaller than the dimensionality of the Inception features. A lower KID score indicates greater visual similarity between real and generated images [47]. KID uses a polynomial kernel $k(x, y) = (\frac{1}{d}x^T \cdot y + 1)^3$ where d is the dimension of the Inception representation vector, and x and y are two feature vectors. The kernel $K(x, y) = k(\phi(x), \phi(y))$ can then be applied as an MMD for input images, where images are mapped into Inception representations through ϕ [48]. Since KID is a distance metric, a value of 0 indicates a perfect match between the distributions of real and generated features. Theoretically, if the two distributions are identical, the distance equals zero. A key advantage of KID lies in its simple and unbiased estimator, which distinguishes it from FID [48].

3) *Multi-Scale Structural Similarity (MS-SSIM)*: MS-SSIM is a multi-scale improvement of SSIM that aims to eliminate aspects of an image that are not crucial for human perception [49], [50]. The calculation of SSIM, which serves as the basis for MS-SSIM, compares three principal components: luminance (intensity), contrast, and structure. It employs basic statistical moments, such as the mean and standard deviation, within local windows to derive a similarity score.

$$MS-SSIM(x, y) = [l_M(x, y)]^{\alpha_M} \cdot \prod_{j=1}^M [c_j(x, y)]^{\beta_j} \cdot [s_j(x, y)]^{\gamma_j} \quad (13)$$

where $l_M(x, y)$ represents the luminance component between images x and y , $c_j(x, y)$ denotes the contrast component at scale j , and $s_j(x, y)$ reflects the structural similarity in scale j . The parameters α_M , β_j , and γ_j define the relative importance of the luminance, contrast, and structure components, respectively. Lastly, M signifies the number of scales employed in the MS-SSIM calculation [51]. The range of MS-SSIM values is between 0 and 1, and images with higher MS-SSIM values are perceived as more similar [52], with 1 indicating perfect similarity. Lower MS-SSIM scores indicate

higher diversity [46], [53]. Unlike metrics such as FID (Fréchet Inception Distance) or KID (Kernel Inception Distance), MS-SSIM does not require a pre-trained Inception network [54]. Therefore, the use of IS, KID, or FID is not recommended, as these metrics inherently utilize weights from the Inception network, which are validated for images similar to those in the ImageNet dataset. Consequently, these metrics cannot be applied to datasets outside ImageNet. MS-SSIM was applied to a fingerprint dataset [54]. Lower FID values indicate superior image quality, and lower MS-SSIM scores reflect increased diversity [53]. Smaller FID and KID values correlate with more realistic images in image translation [55]. Considering that our spectrogram dataset is not part of the ImageNet dataset, we utilized the MS-SSIM score.

V. RESULTS

We identified an unbalanced data distribution, consisting of 3% (HY), 9% (LP), 36% (TC), 8% (TR), and 44% (VT), as shown in Table III. To address this imbalance, we generated new data through various data augmentation techniques, such as Rotation, Jittering, Drifting, Scaling, Flipping, Genetic Algorithms, SpecAugment, and Interpolation. Each data augmentation technique was applied to balance the existing dataset, resulting in a dataset containing 13,430 events (2,686 examples per class). Furthermore, we created balanced datasets consisting solely of augmented data, ensuring an equal number of events, as detailed in Table V. Multiple experiments were designed based on each dataset generated by the respective DA methods (Table VI). For each resulting balanced dataset, a deep learning model was trained via a transfer learning strategy using the pre-trained AlexNet model. The training process involved 20 epochs, a mini-batch size of 8, and an initial learning rate of 0.00005 and the Adam optimization algorithm. The training and validation performance scores for selected experiments are illustrated in Figure 10. The trained models were subsequently evaluated using the original unbalanced dataset of 6,145 events, with the results presented in Table VII. Additionally, confusion matrices for selected experiments are depicted in Figure 11.

A. Real

The original unbalanced dataset, comprising 6,145 events, was utilized, divided into proportions of 80% for training and 20% for testing the model. This experiment aims to highlight the influence of an unbalanced dataset on the accuracy of the probabilistic model.

B. Rotation

The original unbalanced dataset of 6,145 events has been considered and subsequently balanced using rotation as a data augmentation technique. This approach increases the number of examples from underrepresented classes, resulting in a balanced dataset of 13,430 events. This dataset will demonstrate the effectiveness of this data augmentation technique in constructing a classification model. Additionally, a balanced dataset consisting solely of 13,430 augmented events was

TABLE V: Statistics related to the transfer learning over AlexNet pre-trained model.

Dataset	Data type	Data Augmentation	Balanced dataset	Epochs	Total data	Train data	Test data	Training time (min.)
i	Real	No	NO	10	6,145	4,916	1,229	4.2
ii	Real + Augmented	Rotation(5%-25%)	YES	20	13,430	10,744	2,686	18.7
iii	Augmented	Rotation(5%-25%)	YES	20	13,430	10,744	2,686	20.9
iv	Real + Augmented	Jittering(0.2)	YES	20	13,430	10,744	2,686	16.8
v	Augmented	Jittering(0.2)	YES	20	13,430	10,744	2,686	17.2
vi	Real + Augmented	Drifting Drift(0.01-0.1)	YES	20	13,430	10,744	2,686	17.9
vii	Augmented	Drifting Drift(0.01-0.1)	YES	20	13,430	10,744	2,686	20.3
viii	Real + Augmented	Drifting Drift(0.001-0.01)	YES	20	13,430	10,744	2,686	21.7
ix	Augmented	Drifting Drift(0.001-0.01)	YES	20	13,430	10,744	2,686	21.0
x	Real + Augmented	Drifting Drift(0.0001-0.001)	YES	20	13,430	10,744	2,686	18.1
xi	Augmented	Drifting Drift(0.0001-0.001)	YES	20	13,430	10,744	2,686	20.9
xii	Real + Augmented	Genetic Algorithm Segment Size(5s) Crossover Segments(3)	YES	20	13,430	10,744	2,686	16.7
xiii	Augmented	Genetic Algorithm Segment Size(5s) Crossover Segments(3)	YES	20	13,430	10,744	2,686	17.0
xiv	Real + Augmented	Genetic Algorithm Segment Size(5s) Crossover Segments(5)	YES	20	13,430	10,744	2,686	16.6
xv	Augmented	Genetic Algorithm Segment Size(5s) Crossover Segments(5)	YES	20	13,430	10,744	2,686	16.9
xvi	Real + Augmented	Genetic Algorithm Signal Percent(40%) Crossover Segments(10)	YES	20	13,430	10,744	2,686	16.6
xvii	Augmented	Genetic Algorithm Signal Percent(40%) Crossover Segments(10)	YES	20	13,430	10,744	2,686	17.3
xviii	Real + Augmented	Genetic Algorithm Signal Percent(45%) Crossover Segments(10)	YES	20	13,430	10,744	2,686	16.5
xix	Augmented	Genetic Algorithm Signal Percent(45%) Crossover Segments(10)	YES	20	13,430	10,744	2,686	17.1
xx	Real + Augmented	SpecAugment Frequency Percent(0%-10%) Frequency Masks(2) Time Percent(0%-10%) Time Marks(2)	YES	20	13,430	10,744	2,686	16.5
xxi	Augmented	SpecAugment Frequency Percent(0%-10%) Frequency Masks(2) Time Percent(0%-10%) Time Marks(2)	YES	20	13,430	10,744	2,686	17.0
xxii	Real + Augmented	SpecAugment Frequency Percent(0%-15%) Frequency Masks(2) Time Percent(0%-15%) Time Marks(2)	YES	20	13,430	10,744	2,686	16.6
xxiii	Augmented	SpecAugment Frequency Percent(0%-15%) Frequency Masks(2) Time Percent(0%-15%) Time Marks(2)	YES	20	13,430	10,744	2,686	16.9
xxiv	Real + Augmented	Interpolation AE Signal percent(40%-60%)	YES	20	13,430	10,744	2,686	16.6
xxv	Augmented	Interpolation AE Signal percent(40%-60%)	YES	20	13,430	10,744	2,686	17.9

TABLE VI: Performance of the transfer learning experiments over AlexNet pre-trained model.

Dataset	Data augmentation	Accuracy (%)	F1 (%)	Recall (%)	Precision (%)	Cohen Kappa (%)	Class Accuracy (%)				
							HY	LP	TC	TR	VT
i	No	62.8	63.9	62.8	66.7	67.6	38.9	74.8	73.5	33.3	93.4
ii	Rotation(5%-25%)	91.5	91.4	91.5	91.4	89.2	98.4	92.4	83.5	93.3	89.6
iii	Rotation(5%-25%)	94.2	94.0	94.2	94.0	92.6	99.8	96.3	83.1	96.1	95.8
iv	Jittering(0.2)	90.2	90.0	90.2	90.2	87.5	94.0	93.6	77.9	95.9	89.4
v	Jittering(0.2)	91.1	91.1	91.1	91.2	88.8	97.1	95.3	85.2	93.3	84.3
vi	Drifting Drift(0.01-0.1)	84.1	83.5	84.1	83.7	79.6	94.4	90.7	62.1	82.4	90.8
vii	Drifting Drift(0.01-0.1)	87.7	87.6	87.7	87.9	84.6	98.7	88.5	78.1	91.3	81.7
viii	Drifting Drift(0.001-0.01)	90.8	90.6	90.8	90.6	88.3	96.9	95.7	77.7	90.2	93.6
ix	Drifting Drift(0.001-0.01)	93.5	93.4	93.5	93.5	91.8	98.9	95.9	83.0	97.0	92.8
x	Drifting Drift(0.0001-0.001)	93.8	93.5	93.8	93.7	92.0	99.8	100.0	81.0	98.0	90.0
xi	Drifting Drift(0.0001-0.001)	97.6	97.6	97.6	97.6	97.0	100.0	99.0	93.4	100.0	95.6
xii	Genetic Algorithm Segment Size(5s) Crossover Segments(3)	87.8	87.6	87.8	87.7	84.6	95.8	91.5	73.6	91.7	86.7
xiii	Genetic Algorithm Segment Size(5s) Crossover Segments(3)	86.4	86.2	86.4	86.3	82.8	92.6	90.7	74.6	85.6	88.6
xiv	Genetic Algorithm Segment Size(5s) Crossover Segments(5)	86.4	86.1	86.4	86.2	82.7	94.6	88.5	72.0	86.0	91.0
xv	Genetic Algorithm Segment Size(5s) Crossover Segments(5)	85.8	85.5	85.8	85.9	81.9	90.4	94.4	75.1	77.3	91.8
xvi	Genetic Algorithm Signal Percent(40%) Crossover Segments(10)	84.9	84.7	84.9	84.7	80.8	88.1	91.5	74.1	80.0	90.8
xvii	Genetic Algorithm Signal Percent(40%) Crossover Segments(10)	83.0	82.9	83.0	83.0	78.6	89.2	83.1	77.2	73.9	91.8
xviii	Genetic Algorithm Signal Percent(45%) Crossover Segments(10)	83.7	83.4	83.7	83.8	79.3	94.0	84.3	75.0	73.2	91.8
xix	Genetic Algorithm Signal Percent(45%) Crossover Segments(10)	83.2	83.2	83.2	83.3	78.8	89.5	86.0	75.3	76.9	88.2
xx	SpecAugment Frequency Percent(0%-10%) Frequency Masks(2) Time Percent(0%-10%) Time Marks(2)	90.5	90.2	90.5	90.2	87.9	98.2	95.5	77.9	90.2	90.4
xxi	SpecAugment Frequency Percent(0%-10%) Frequency Masks(2) Time Percent(0%-10%) Time Marks(2)	91.7	91.5	91.7	91.5	89.4	97.5	94.8	83.5	89.5	93.2
xxii	SpecAugment Frequency Percent(0%-15%) Frequency Masks(2) Time Percent(0%-15%) Time Marks(2)	88.6	88.6	88.6	88.5	85.7	94.9	86.8	80.5	89.5	91.4
xxiii	SpecAugment Frequency Percent(0%-15%) Frequency Masks(2) Time Percent(0%-15%) Time Marks(2)	86.6	86.4	86.6	86.4	83.1	96.0	85.6	75.8	87.2	88.2
xxiv	Interpolation AE Signal percent(40%-60%)	87.5	87.4	87.5	87.6	84.1	89.3	90.9	76.9	89.3	91.0
xxv	Interpolation AE Signal percent(40%-60%)	93.1	93.1	93.1	93.3	91.3	97.6	96.5	83.7	96.5	91.4

generated, with 2,686 examples for each class, for comparative purposes. The rotation parameters employed ranged from 5% to 25%, utilizing random rotation axes (start, center, and end) without repetition.

C. Jittering

The original unbalanced dataset, comprising 6,145 events, was utilized, and data augmentation techniques were applied to achieve balance by increasing the number of examples from underrepresented classes. This process resulted in a balanced dataset of 13,430 events. The parameters employed include Gaussian noise, distributed as $N(\mu = 0, \sigma = 0.2)$. Additionally, a balanced dataset of 13,430 events was generated entirely from augmented data.

D. Drifting

The original unbalanced dataset of 6,145 events was considered, and the data augmentation technique was applied in a

manner that balanced the dataset by increasing examples from underrepresented classes, yielding a total of 13,430 events. The parameters utilized include random values sampled from defined real intervals $[a, b]$. Similarly, a balanced dataset consisting exclusively of 13,430 augmented events was generated.

E. Genetic Algorithm

Utilizing the data augmentation technique of Genetic Algorithms, a balanced dataset of 13,430 augmented events was generated, comprising 2,686 examples for each class. Various values for the signal proportion parameters of the parent events, as well as the number of crossover segments, were considered for generating the new dataset. This dataset aims to demonstrate the effectiveness of this data augmentation technique in developing the model. This data augmentation technique includes two variants:

- *Without Time Adjustment:* In this variant, the parents are segmented into the specified number of crossover

TABLE VII: Test of trained models over the real dataset of 6,145 events.

Dataset	Data augmentation	Accuracy (%)	F1 (%)	Recall (%)	Precision (%)	Cohen Kappa (%)	Class Accuracy (%)				
							HY	LP	TC	TR	VT
i	No	-	-	-	-	-	-	-	-	-	-
ii	Rotation(5%-25%)	96.6	96.2	96.6	95.9	95.7	97.7	95.3	97.0	94.9	98.1
iii	Rotation(5%-25%)	93.7	88.9	93.7	85.5	86.9	100.0	96.0	82.5	93.8	96.0
iv	Jittering(0.2)	96.2	96.0	96.2	95.8	95.0	94.8	96.7	95.8	96.2	97.5
v	Jittering(0.2)	93.9	87.3	93.9	82.9	86.2	99.5	97.1	86.0	95.8	91.3
vi	Drifting Drift(0.01-0.1)	91.5	89.4	91.5	88.1	90.0	93.0	95.5	90.0	81.5	97.6
vii	Drifting Drift(0.01-0.1)	66.6	53.5	66.6	55.2	48.3	69.5	96.5	42.0	56.5	68.4
viii	Drifting Drift(0.001-0.01)	95.0	95.4	95.0	95.8	94.7	93.0	97.1	94.9	91.3	98.8
ix	Drifting Drift(0.001-0.01)	93.7	86.7	93.7	82.4	85.5	98.1	97.4	83.4	97.7	91.8
x	Drifting Drift(0.0001-0.001)	98.3	96.7	98.3	95.3	96.0	99.5	100.0	95.5	98.5	97.8
xi	Drifting Drift(0.0001-0.001)	97.2	93.8	97.2	91.0	92.9	100.0	98.3	93.3	99.6	94.9
xii	Genetic Algorithm Segment Size(5s) Crossover Segments(3)	95.9	93.7	95.9	91.8	93.8	96.7	97.6	94.1	94.3	97.0
xiii	Genetic Algorithm Segment Size(5s) Crossover Segments(3)	82.7	73.6	82.7	70.0	71.8	86.9	93.4	65.2	78.3	89.8
xiv	Genetic Algorithm Segment Size(5s) Crossover Segments(5)	95.2	93.6	95.2	92.2	93.6	95.3	96.7	93.2	92.4	98.3
xv	Genetic Algorithm Segment Size(5s) Crossover Segments(5)	75.2	67.3	75.2	64.9	64.3	77.5	91.2	57.3	61.4	88.8
xvi	Genetic Algorithm Signal Percent(40%) Crossover Segments(10)	93.3	94.0	93.3	94.9	93.8	90.6	98.1	95.5	84.1	98.3
xvii	Genetic Algorithm Signal Percent(40%) Crossover Segments(10)	81.4	74.2	81.4	70.7	72.4	85.4	83.9	67.9	78.6	91.1
xviii	Genetic Algorithm Signal Percent(45%) Crossover Segments(10)	94.0	94.0	94.0	94.2	93.8	95.3	94.8	95.1	86.4	98.5
xix	Genetic Algorithm Signal Percent(45%) Crossover Segments(10)	80.1	68.1	80.1	64.9	65.8	88.7	92.5	56.9	76.6	85.9
xx	SpecAugment Frequency Percent(0%-10%) Frequency Masks(2) Time Percent(0%-10%) Time Marks(2)	96.7	96.4	96.7	96.1	95.4	98.6	98.6	95.8	92.4	98.2
xxi	SpecAugment Frequency Percent(0%-10%) Frequency Masks(2) Time Percent(0%-10%) Time Marks(2)	95.4	91.1	95.4	87.8	89.8	100.0	97.2	88.1	96.6	95.1
xxii	SpecAugment Frequency Percent(0%-15%) Frequency Masks(2) Time Percent(0%-15%) Time Marks(2)	95.5	95.7	95.5	95.9	94.9	97.7	90.5	96.3	94.9	98.4
xxiii	SpecAugment Frequency Percent(0%-15%) Frequency Masks(2) Time Percent(0%-15%) Time Marks(2)	93.6	87.5	93.6	83.7	85.2	100.0	95.1	82.8	97.5	92.4
xxiv	Interpolation AE Signal percent(40%-60%)	90.4	91.8	90.4	93.4	92.0	82.2	90.5	94.7	86.4	98.1
xxv	Interpolation AE Signal percent(40%-60%)	80.7	71.0	80.7	67.5	67.8	88.7	85.8	60.4	80.7	88.1

segments, with crossover occurring at defined times. The parameters used include three crossover segments, each lasting 5 seconds.

- *With Time Adjustment*: In this variant, both the number of crossover segments and the signal proportion for the crossover are defined. The segment times are calculated proportionally for both parents.

F. SpecAugment

Using the SpecAugment data augmentation technique, a balanced dataset of 13,430 augmented events was generated, comprising 2,686 examples for each class. This dataset aims to demonstrate the effectiveness of this data augmentation technique in developing the model. To generate the new dataset, different values for the maximum percentage parameters of each time and frequency mask were considered, along with the number of masks applied. The parameters employed include two time and frequency masks, each covering 10% of the signal.

G. Interpolation AEs

This technique was utilized to generate a balanced dataset of 13,430 purely augmented events. For this technique, a randomly selected parameter $\alpha \in [0.4, 0.6]$ was used in conjunction with a specifically designed convolutional autoencoder to reconstruct the interpolated signal, as illustrated in Figure 9.

The experimental evidence underscores the importance of data augmentation techniques in enhancing the precision of deep learning models. Models trained without these data augmentation techniques exhibited low precision, as illustrated in Table V. To compare our models with the state-of-the-art, we considered performance metrics, including

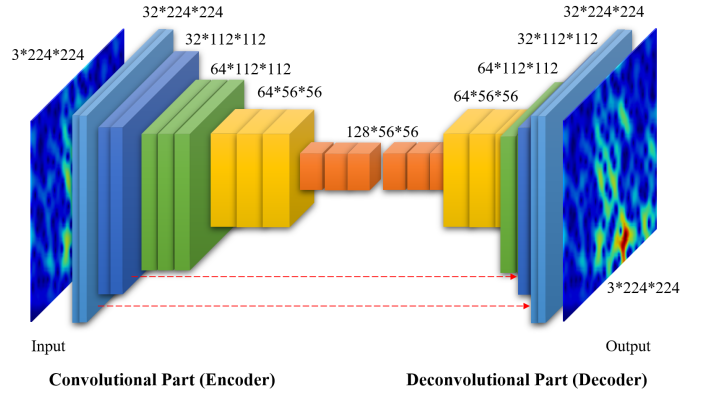


Fig. 9: Convolutional AutoEncoder of spectrograms.

accuracy, precision, recall, specificity, and F1 score.

H. Diversity and Similarity Metrics

The diversity and similarity metrics are calculated by comparing two datasets and the same event type. This metric is called an *internal* diversity and similarity metric if only the internal data of the dataset are considered for the comparison (Table IX). Depending on the dataset type, the following cases apply:

- *Real + augmented*: Since the VT event has sufficient examples and does not require augmented data to balance the dataset, this metric is not calculated for this specific event due to the lack of comparison data. For the other events, the comparison is made between real and augmented data.
- *Augmented*: Since all data is augmented, it is divided into two distinct subsets. The subsequent analysis and com-

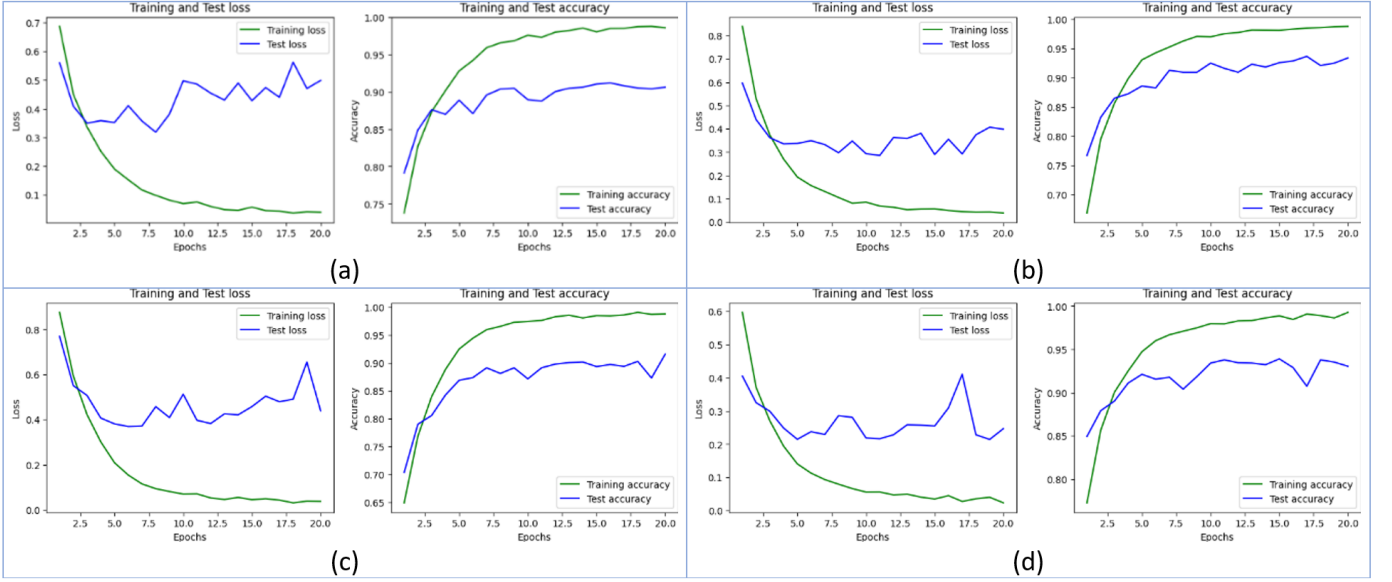


Fig. 10: Training and validation performance scores using transfer learning with AlexNet. Loss and accuracy in training and testing of the experiment on the dataset: a) Drifting Drift(0.001-0.01)(Real+Augmented)(viii), b) Drifting Drift(0.001-0.01)(Augmented)(ix), c) SpecAugment Frequency Percent(0%-10%) Frequency Masks(2) Time Percent(0%-10%) Time Marks(2)(Augmented)(xxi), and d) Interpolation AE Signal Percent(40%-60%)(Augmented)(xxv).

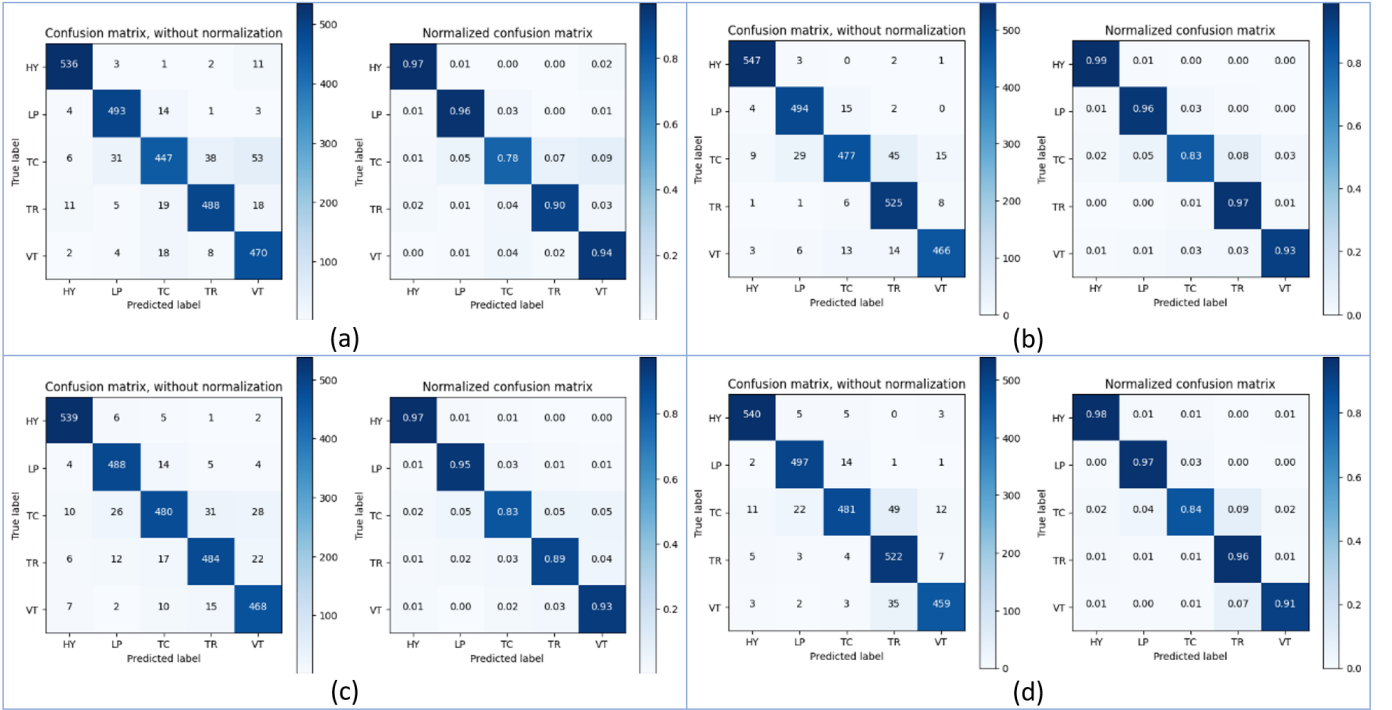


Fig. 11: Normalized and non-normalized confusion matrix of model training using transfer learning with AlexNet on the dataset: a) Drifting Drift(0.001-0.01)(Real+Augmented)(viii), b) Drifting Drift(0.001-0.01)(Augmented)(ix), c) SpecAugment Frequency Percent(0%-10%) Frequency Masks(2) Time Percent(0%-10%) Time Marks(2)(Augmented)(xxi), and d) Interpolation AE Signal Percent(40%-60%)(Augmented)(xxv).

parison are then conducted between these two resulting subsets.

It is called an *external* diversity and similarity metric if the comparison is made between two datasets, event by event. The datasets considered for this comparison are the study dataset

and the real dataset (Table X).

I. Performace Score

By generating a simple score that accounts for model training accuracy (higher value is better), model test accuracy on real

TABLE VIII: Relationship between Data Augmentation technique and Dataset.

Code	Data augmentation technique	Datasets	
		Real + Augmented	Augmented
N	Real dataset	i	
R	Rotation (5%-25%)	ii	iii
J	Jittering (0.2)	iv	v
D	Drifting (0.01-0.1)	vi	vii
DD	Drifting (0.001-0.01)	viii	ix
DDD	Drifting (0.0001-0.001)	x	xi
G	Genetic Algorithm Segment Size(5s) Crossover Segments(3)	xii	xiii
GG	Genetic Algorithm Segment Size(5s) Crossover Segments(5)	xiv	xv
H	Genetic Algorithm Signal Percent(40%) Crossover Segments(10)	xvi	xvii
HH	Genetic Algorithm Signal Percent(45%) Crossover Segments(10)	xviii	xix
S	SpecAugment Frequency Percent(0%-10%) Frequency Masks(2) Time Percent(0%-10%) Time Marks(2)	xx	xxi
SS	SpecAugment Frequency Percent(0%-15%) Frequency Masks(2) Time Percent(0%-15%) Time Marks(2)	xxii	xxiii
I	Interpolation AE Signal percent(40%-60%)	xxiv	xv

TABLE IX: Internal Similarity and Diversity metrics for experimental datasets.

Dataset	Data augmentation technique	Fr�chet Inception Distance (FID)					Kernel Inception Distance (KID)					Multiscale Structural Similarity Index (MS-SSIM)							
		HY	LP	TC	TR	VT	FID	HY	LP	TC	TR	VT	KID	HY	LP	TC	TR	VT	MS
i	No	0.16	0.10	0.02	0.05	0.02	0.03	0.14	-0.75	3.78	0.31	-0.23	1.21	0.34	0.41	0.33	0.29	0.34	0.34
ii	Rotation(5%-25%)	0.70	0.61	0.50	0.45		0.56	4.81	-1.40	4.41	2.56		2.59	0.38	0.42	0.36	0.29		0.36
iii	Rotation(5%-25%)	0.03	0.01	0.04	0.02	0.03	0.02	1.62	0.15	1.16	-1.03	2.34	0.85	0.36	0.42	0.36	0.33	0.35	0.37
iv	Jittering(0.2)	10.09	5.46	7.65	8.88		8.02	56.99	18.80	40.98	48.72		41.37	0.28	0.36	0.27	0.26		0.29
v	Jittering(0.2)	0.01	0.01	0.02	0.02	0.03	0.02	-2.50	-2.35	3.38	0.24	-2.77	-0.80	0.26	0.33	0.26	0.21	0.23	0.26
vi	Drifting Drift(0.01-0.1)	34.80	30.48	35.93	30.01		32.81	102.53	71.83	89.02	83.60		86.75	0.40	0.46	0.42	0.35		0.41
vii	Drifting Drift(0.01-0.1)	0.01	0.02	0.01	0.01	0.01	0.01	-0.06	-0.10	0.29	0.51	-0.11	0.11	0.80	0.81	0.81	0.76	0.71	0.78
viii	Drifting Drift(0.001-0.01)	1.50	1.23	2.69	0.84		1.56	3.25	3.61	3.84	1.01		2.93	0.37	0.42	0.38	0.31		0.37
ix	Drifting Drift(0.001-0.01)	0.03	0.01	0.02	0.03	0.01	0.02	1.53	-1.02	0.30	1.14	-0.80	0.23	0.42	0.48	0.43	0.37	0.36	0.41
x	Drifting Drift(0.0001-0.001)	0.02	0.02	0.03	0.02		0.02	-4.42	-4.68	-3.96	0.79		-0.73	0.35	0.40	0.33	0.29		0.34
xi	Drifting Drift(0.0001-0.001)	0.01	0.01	0.02	0.01	0.01	0.01	-5.60	-1.65	2.42	3.00	1.74	-0.02	0.35	0.40	0.32	0.31	0.34	0.34
xii	Genetic Algorithm Segment Size(5s) Crossover Segments(3)	0.89	0.41	0.39	0.28		0.49	2.36	2.98	-1.94	2.55		1.49	0.38	0.40	0.35	0.30		0.36
xiii	Genetic Algorithm Segment Size(5s) Crossover Segments(3)	0.02	0.01	0.00	0.01	0.01	0.01	0.45	-2.49	-0.40	-2.06	2.54	-0.39	0.35	0.40	0.37	0.30	0.36	0.36
xiv	Genetic Algorithm Segment Size(5s) Crossover Segments(5)	1.65	0.83	0.76	1.03		1.07	1.51	4.59	2.90	1.29		2.57	0.35	0.41	0.35	0.30		0.35
xv	Genetic Algorithm Segment Size(5s) Crossover Segments(5)	0.02	0.01	0.01	0.02	0.01	0.01	1.06	-0.48	-0.32	-2.37	-1.77	-0.77	0.35	0.40	0.37	0.33	0.40	0.37
xvi	Genetic Algorithm Signal Percent(40%) Crossover Segments(10)	0.90	0.63	0.55	0.41		0.63	3.99	1.91	0.08	4.25		2.56	0.36	0.41	0.35	0.30		0.35
xvii	Genetic Algorithm Signal Percent(40%) Crossover Segments(10)	0.02	0.02	0.04	0.01	0.01	0.02	-0.28	-3.37	-0.14	1.87	-0.70	-0.52	0.34	0.41	0.32	0.29	0.34	0.34
xviii	Genetic Algorithm Signal Percent(45%) Crossover Segments(10)	0.97	0.67	0.73	0.36		0.68	4.43	-0.28	3.54	0.97		2.16	0.35	0.41	0.33	0.30		0.35
xix	Genetic Algorithm Signal Percent(45%) Crossover Segments(10)	0.01	0.00	0.00	0.01	0.01	0.01	0.91	0.75	-0.73	2.75	0.86	0.91	0.31	0.41	0.35	0.30	0.34	0.34
xx	SpecAugment Frequency Percent(0%-10%) Frequency Masks(2) Time Percent(0%-10%) Time Marks(2)	1.53	1.48	1.85	1.21		1.52	6.23	4.41	6.33	2.06		4.76	0.37	0.41	0.33	0.30		0.35
xxi	SpecAugment Frequency Percent(0%-10%) Frequency Masks(2) Time Percent(0%-10%) Time Marks(2)	0.00	0.01	0.02	0.01	0.01	0.01	-1.21	2.46	-0.33	-1.04	1.03	0.18	0.34	0.40	0.35	0.31	0.34	0.35
xxii	SpecAugment Frequency Percent(0%-15%) Frequency Masks(2) Time Percent(0%-15%) Time Marks(2)	3.84	3.45	3.08	3.22		3.40	16.73	11.34	7.82	6.15		10.51	0.35	0.41	0.33	0.31		0.35
xxiii	SpecAugment Frequency Percent(0%-15%) Frequency Masks(2) Time Percent(0%-15%) Time Marks(2)	0.01	0.01	0.01	0.00	0.01	0.01	0.31	1.29	0.47	0.72	-0.61	0.44	0.35	0.40	0.36	0.32	0.39	0.37
xxiv	Interpolation AE Signal percent(40%-60%)	2.31	2.46	2.18	1.88		2.21	9.48	3.04	4.03	7.12		5.92	0.38	0.44	0.36	0.33		0.38
xxv	Interpolation AE Signal percent(40%-60%)	0.01	0.01	0.01	0.01	0.01	0.01	-0.22	0.62	-0.89	-0.44	0.03	-0.18	0.43	0.48	0.41	0.37	0.41	0.42

data (higher value is better), and the MS-SSIM metric value (lower value is better, indicating greater diversity), we define the Performance Score (PS) as follows:

$$PS = \frac{A_{tr} + 2 \cdot A_{te} + (1 - MS-SSIM)}{4} \quad (14)$$

where A_{tr} is the model training accuracy, A_{te} is the model test accuracy on real data, and $MS-SSIM$ is the metric value. All components, A_{tr} , A_{te} and $MS-SSIM$, are defined within the range $[0, 1]$. The model test accuracy (A_{te}) was assigned a higher weight (2x) due to its direct relationship with the model’s generalization ability. For the calculation of this score, the *external* MS-SSIM metric was selected because it considers the entire real dataset for its computation.

VI. DISCUSSION

The various experiments conducted in this research highlight the performance of the probabilistic model on unbalanced datasets. Specifically, Real dataset (i) produced a model biased toward the LP, TC, and VT events, as shown in Table VI. The results indicate that balanced datasets yield higher-performing models, as they expand the examples within the event feature space. The models were subsequently tested on real data, demonstrating competitive classification levels in

most cases. Transfer learning facilitated the development of deep learning models in a reduced timeframe, as evidenced in Table VI. In tables VI and VII, it is evident that the Drifting data augmentation technique achieved the best results across almost all pre-trained models. This success may be attributed to the Drifting technique’s ability to effectively cover the data space. The most notable result is from Drifting (Augmented)(xi), which, with a drift range of 0.0001-0.001, achieved an accuracy of 97.6% over an entirely augmented dataset. An additional noteworthy result is the Rotation (Augmented)(iii) technique, which, with a range of 5%-25%, achieved an accuracy of 94.2% on another augmented dataset. In both cases, the F1 score metrics closely correspond to their respective accuracies. Although both models were derived from entirely augmented data, they also performed well in real data tests, achieving accuracies of 97.2% for Drifting (Augmented)(xi) and 93.7% for Rotation (Augmented)(iii) (Table VII). However, in tests on real data, the Drifting (Real+Augmented)(x) model achieved the highest accuracy (98.3%), using both real and augmented data. Furthermore, this dataset (Real+Augmented)(x), generated by Drifting in the range of 0.0001 to 0.001, achieved an accuracy of 93.8% during model training. Then the dataset of augmented

TABLE X: External Similarity and Diversity metrics for experimental datasets.

Dataset	Data augmentation technique	Multiscale Structural Similarity Index (MS-SSIM)					
		HY	LP	TC	TR	VT	MS-SSIM
i	No	1.00	1.00	1.00	1.00	1.00	1.00
ii	Rotation(5%-25%)	0.37	0.41	0.33	0.32	1.00	0.53
iii	Rotation(5%-25%)	0.39	0.41	0.34	0.32	0.37	0.37
iv	Jittering(0.2)	0.26	0.37	0.32	0.25	1.00	0.49
v	Jittering(0.2)	0.30	0.34	0.27	0.24	0.26	0.28
vi	Drifting Drift(0.01-0.1)	0.44	0.45	0.34	0.33	1.00	0.55
vii	Drifting Drift(0.01-0.1)	0.44	0.47	0.39	0.34	0.38	0.40
viii	Drifting Drift(0.001-0.01)	0.38	0.42	0.33	0.30	1.00	0.53
ix	Drifting Drift(0.001-0.01)	0.36	0.43	0.35	0.31	0.33	0.35
x	Drifting Drift(0.0001-0.001)	0.38	0.41	0.33	0.29	1.00	0.53
xi	Drifting Drift(0.0001-0.001)	0.32	0.40	0.33	0.30	0.35	0.34
xii	Genetic Algorithm Segment Size(5s) Crossover Segments(3)	0.43	0.41	0.33	0.30	1.00	0.54
xiii	Genetic Algorithm Segment Size(5s) Crossover Segments(3)	0.37	0.40	0.32	0.30	0.35	0.35
xiv	Genetic Algorithm Segment Size(5s) Crossover Segments(5)	0.36	0.40	0.33	0.30	1.00	0.52
xv	Genetic Algorithm Segment Size(5s) Crossover Segments(5)	0.40	0.40	0.34	0.28	0.38	0.36
xvi	Genetic Algorithm Signal Percent(40%) Crossover Segments(10)	0.35	0.40	0.32	0.30	1.00	0.52
xvii	Genetic Algorithm Signal Percent(40%) Crossover Segments(10)	0.37	0.40	0.33	0.30	0.34	0.34
xviii	Genetic Algorithm Signal Percent(45%) Crossover Segments(10)	0.36	0.40	0.33	0.29	1.00	0.52
xix	Genetic Algorithm Signal Percent(45%) Crossover Segments(10)	0.32	0.41	0.33	0.30	0.35	0.34
xx	SpecAugment Frequency Percent(0%-10%) Frequency Masks(2) Time Percent(0%-10%) Time Marks(2)	0.39	0.40	0.34	0.30	1.00	0.53
xxi	SpecAugment Frequency Percent(0%-10%) Frequency Masks(2) Time Percent(0%-10%) Time Marks(2)	0.38	0.40	0.33	0.29	0.35	0.35
xxii	SpecAugment Frequency Percent(0%-15%) Frequency Masks(2) Time Percent(0%-15%) Time Marks(2)	0.35	0.41	0.33	0.31	1.00	0.53
xxiii	SpecAugment Frequency Percent(0%-15%) Frequency Masks(2) Time Percent(0%-15%) Time Marks(2)	0.36	0.40	0.34	0.31	0.37	0.36
xxiv	Interpolation AE Signal percent(40%-60%)	0.37	0.44	0.34	0.32	1.00	0.54
xxv	Interpolation AE Signal percent(40%-60%)	0.37	0.44	0.35	0.33	0.39	0.38

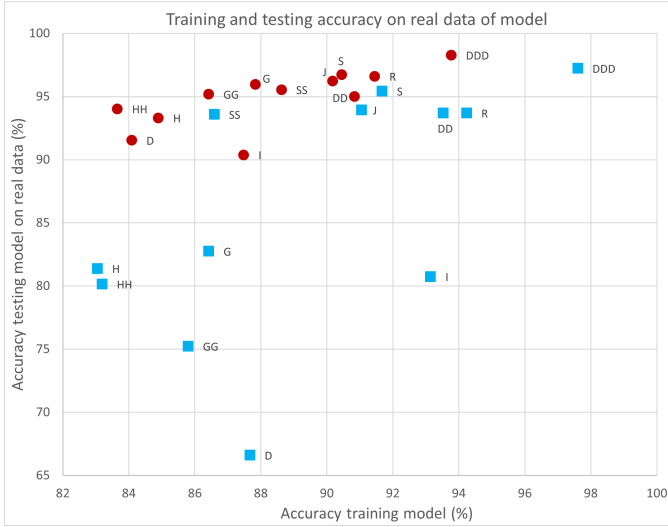


Fig. 12: Accuracy of model training and testing on real data. The green shape indicates real dataset with 62.8% of accuracy (which is not visible because the segment with accuracy $\geq 82\%$ is displayed), red circles indicate datasets with real and augmented data, and turquoise squares are entirely augmented datasets. The codes are described in Table VIII.

data (ix) generated by Drifting in the range 0.001-0.01, reached an accuracy of 93.5%. Then we have the augmented data dataset (xxv) generated by Autoencoder Interpolation

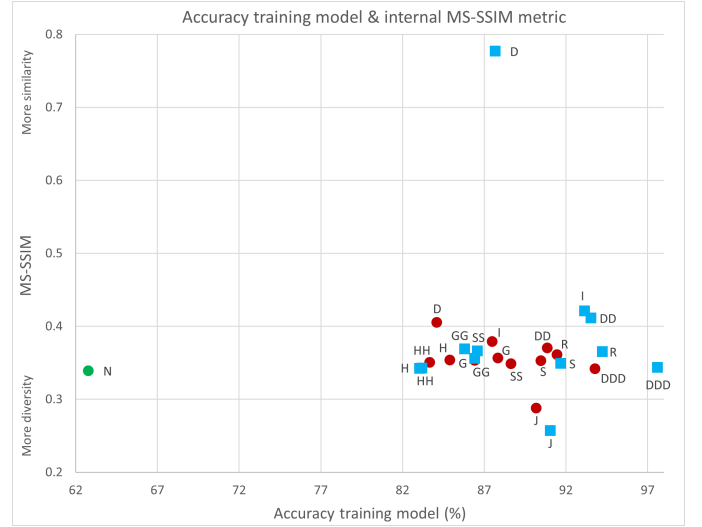


Fig. 13: Accuracy of model training and internal MS-SSIM metric. The green shape indicates real dataset, red circles indicate datasets with real and augmented data, and turquoise squares are entirely augmented datasets. The codes are described in Table VIII.

in the range 40%-60%, it obtained an accuracy of 93.1% (Table VI). These models also show good performance in tests with real data, with accuracies of 98.3% for Drifting (Real+Augmented)(x), 93.7% for Drifting (Augmented)(ix)

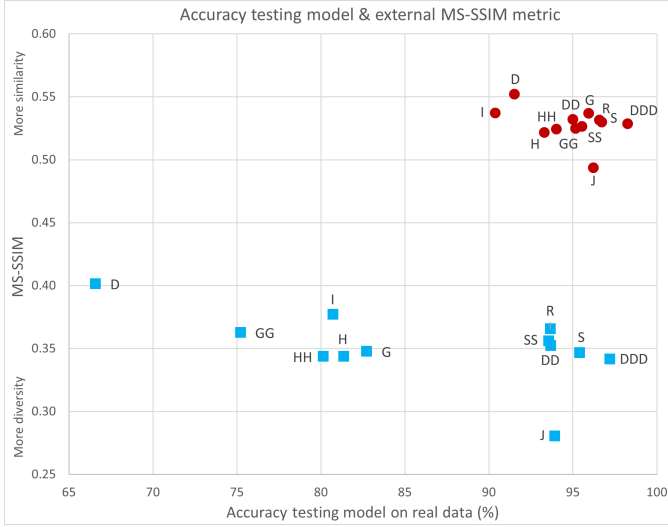


Fig. 18: Accuracy of model testing and external MS-SSIM metric. The red circles indicate datasets with real and augmented data, and turquoise squares are entirely augmented datasets. The codes are described in Table VIII.

(Augmented)(v) and (Real+Augmented)(iv), with MS-SSIM values of 0.26 and 0.29, respectively. The dataset with the highest similarity and lowest diversity is Drifting (0.01-0.1)(Augmented)(vii), which has an MS-SSIM index of 0.78. This is followed by Interpolation (Augmented)(xxv) and Drifting (Real+Augmented)(vi), with MS-SSIM values of 0.42 and 0.41, respectively. Figure 13 illustrates the concentration of points representing the MS-SSIM similarity metric versus the accuracy of model training. Notably, two distant points are evident: the dataset comprising only real data (i), which has low accuracy due to being unbalanced, and the Drifting dataset (Augmented)(vii) with a range of 0.01-0.1, which shows higher similarity. Zooming in on this in Figure 14, it is clear that almost all datasets demonstrate similar levels of diversity, including the real dataset; however, the augmented Jittering dataset (v) shows greater diversity. When comparing datasets composed only of augmented data with those containing both real and augmented data (using the same data augmentation method), the former, in addition to presenting greater or similar diversity, lead to superior training performance, especially with Jittering (Augmented)(v), SpecAugment (Augmented)(xxi) and Rotation (Augmented)(iii). In Figure 15, certain augmented and real datasets demonstrate better diversity and testing accuracy values compared to their counterparts within the same DA method. Notable examples include Drifting (x, xi), Rotation (ii, iii), and SpecAugment (xxii, xxiii).

Regarding the *external* similarity and diversity metrics, Table X shows that the Real dataset exhibits perfect similarity (Figure 16). This result is expected since the dataset is being compared against itself. Figure 17 illustrates the distribution of points according to the MS-SSIM metric plotted against the model's training accuracy. The point corresponding to the Real dataset (i) is noticeable, displaying low accuracy due to

its imbalance and perfect similarity. Two distinct groups are also observable:

- The first group comprises datasets that combine real and augmented data, exhibiting MS-SSIM values in the range of 0.5 to 0.55. Two cases fall outside of this range: the Drifting dataset (Real+Augmented)(vi), which shows the greatest similarity with an MS-SSIM value of 0.552, and the Jittering dataset (Real+Augmented)(iv), which shows the greatest diversity for this group, with a value of 0.49.
- The second group corresponds to the fully augmented datasets, which demonstrate greater diversity than the first group. The MS-SSIM value for this group ranges from 0.3 to 0.4. Two cases fall outside this range: the Drifting dataset (Augmented)(vii), with an MS-SSIM value of 0.401, representing the least diversity within the group, and the Jittering dataset (Augmented)(v), with a value of 0.28, which shows the greatest diversity in this group and overall.

In Figure 18, a similar configuration of two groups is observed, with one group being more concentrated than the other. The most compact group corresponds to the real and augmented datasets, which exhibit less diversity than the second group.

Regarding model performance, the Performance Score (PS), which incorporates the model training accuracy, the model testing accuracy on real data, and the MS-SSIM metric, should be examined. Figure 19 shows that models trained on the Drifting (Augmented)(xi), Jittering (Augmented)(v), SpecAugment (Augmented)(xxi), and Drifting (Augmented)(ix) fully augmented datasets exhibit the highest PS values, making them the most relevant in terms of the trade-off between model accuracy and dataset diversity. In contrast, models with low PS values, such as the Drifting fully augmented dataset (Augmented)(vii) and the real event dataset (i), are negatively impacted by their lower diversity and imbalanced nature, respectively. This analysis conclusively demonstrates the extent to which data augmentation techniques and the inherent diversity of the dataset contributes to model performance.

VII. CONCLUSION

In this work, several data augmentation techniques are applied to the classification of seismic-volcanic signals. Initially, these techniques were employed to balance the datasets and subsequently to generate fully augmented datasets. Deep learning classification models were then trained on these datasets, utilizing transfer learning with the pre-trained AlexNet model. The performance of the trained classification models was evaluated, revealing that data augmentation techniques enhance model performance. It was observed that the trained classification models, in conjunction with the considered data augmentation methods, are competitive with the baselines presented in previous research. Additionally, the following results were achieved:

- The integration of data from multiple stations on the Lascar volcano facilitated the construction of more generalizable probabilistic models, thereby mitigating biases

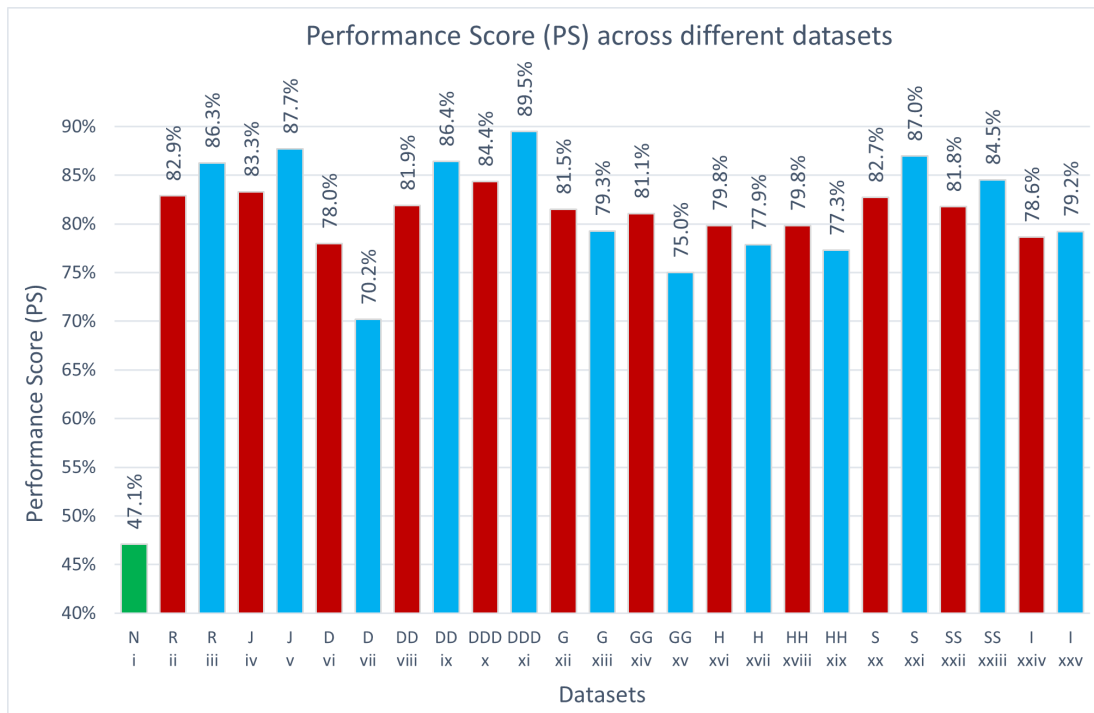


Fig. 19: Performance Score (PS) values on the different datasets (higher is better). The **green** bar indicates real dataset, the **red** bars indicate datasets with real and augmented data, and **turquoise** bars indicate datasets with only augmented data. The codes are described in Table VIII.

associated with individual stations through various data augmentation techniques.

- Data augmentation techniques significantly improve the performance metrics of deep learning models, as they enable the generation of additional examples within the feature space of volcanic seismic events.
- Similarity and diversity metrics were considered to analyze these characteristics in the datasets generated by the various data augmentation techniques presented, in addition to the performance of the models.
- The presented data augmentation techniques generate data with greater diversity, and in several cases, models with better performance.
- Transfer learning significantly accelerated the generation of deep learning models compared to training from scratch.

The findings of this work can be used to improve the performance of deep learning models in seismic-volcanic signal recognition, considering the data augmentation techniques presented. Furthermore, the results indicate that models trained exclusively on augmented datasets show excellent performance when tested with real data, in addition to exhibiting diversity. This finding represents a substantial contribution to seismic-volcanic signal recognition. A repository is available on GitHub for data verification at: <https://github.com/franznet/daseismicsignals>.

VIII. FUTURE WORK

Future research could focus on optimizing the parameters of existing data augmentation methods to improve model

performance. Furthermore, evaluating the effectiveness of these techniques on different types of seismic events and with varying levels of noise would provide deeper insight into their applicability.

The Performance Score was calculated based on three components: the model training accuracy, the model test accuracy on real data, and the MS-SSIM diversity of the dataset. In our initial calculation (Eq. 14), specific constants were defined to facilitate model comparison. However, a hyperparameter could be incorporated to appropriately weigh these components when calculating the score.

To improve model performance, it is essential to quantify the relationship between model accuracy and the diversity generated by the data augmentation method. Specifically, this requires determining the gain in accuracy relative to the variation in dataset diversity.

Exploring a methodology to formulate conditions that determine whether a data augmentation technique increases dataset diversity to safe levels without compromising class separation.

Integrating these data augmentation strategies with other machine learning approaches, such as ensemble learning or reinforcement learning, could lead to the development of more robust and accurate models for seismic-volcanic signal recognition.

Finally, a large dataset could be generated using various data augmentation techniques from different channels, stations, and volcanoes.

Data availability. The data are registered by <https://doi.org/10.5281/Zenodo.6001869> (Salazar et al., 2022). These can also be found in the link: <https://zenodo.org/record/6001870#.YkGXKffQ-XI> (last access: 24 February 2023).

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